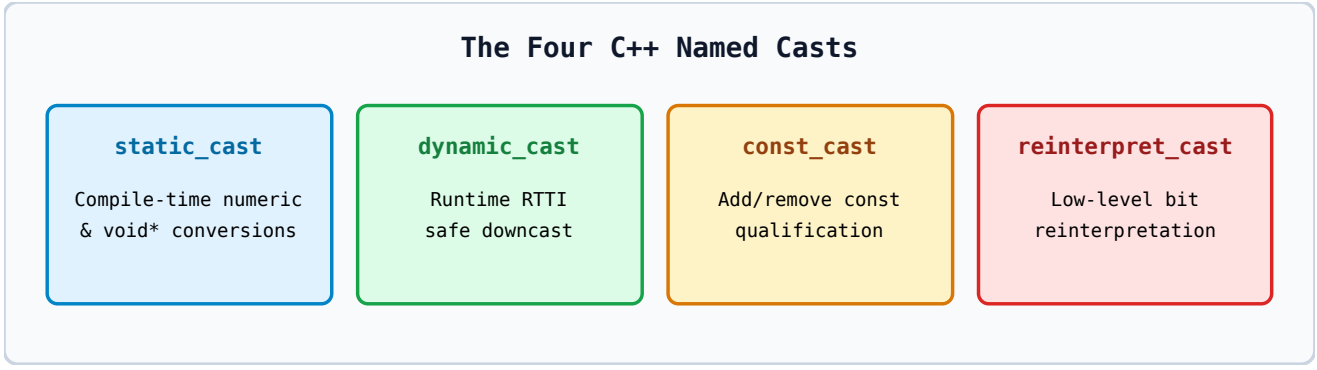
20 TYPE CONVERSION AND CASTING • THEORY

02 the four cpp named casts

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

02 — The Four C++ Named Casts (`static_cast`, `dynamic_cast`, `const_cast`, `reinterpret_cast`)



C++ Named Casts

Named Cast	Primary Purpose	Checked When?
`static_cast<T>(expr)`	Well-defined numeric conversions (`double` to `int`), `void*` to `T*` , related class hierarchy casts	Compile time
`dynamic_cast<T>(expr)`	Safe runtime downcasting (`Base*` to `Derived*`) in polymorphic classes (requires `virtual` function in `Base`)	**Runtime (RTTI)**
`const_cast<T>(expr)`	Adding or removing `const` / `volatile` qualification	Compile time
`reinterpret_cast<T>(expr)`	Low-level bit/address reinterpretation between unrelated pointer types	Compile time